PH.45320 **FALL 2026**

Static Cosmology

Lecture 12: Optical Deflection and Medium Maps

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Overview

Deflection measurements are inverted for a projected medium map with explicit point-spread-function corrections.

1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$\alpha_{\perp} = \gamma \int \nabla_{\perp} \Phi ds / c^2$$

2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$g_{obs} = g_{medium} + e_{intrinsic} + e_{PSF}$$

3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$\Sigma_* = argmin_{\Sigma}(\|K\Sigma - g\|_C^2 + \lambda_R \|L\Sigma\|^2)$$

4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Jeong, ch. 12
- Optical Map Standard OMS-8
- Extinction Atlas geometry appendix